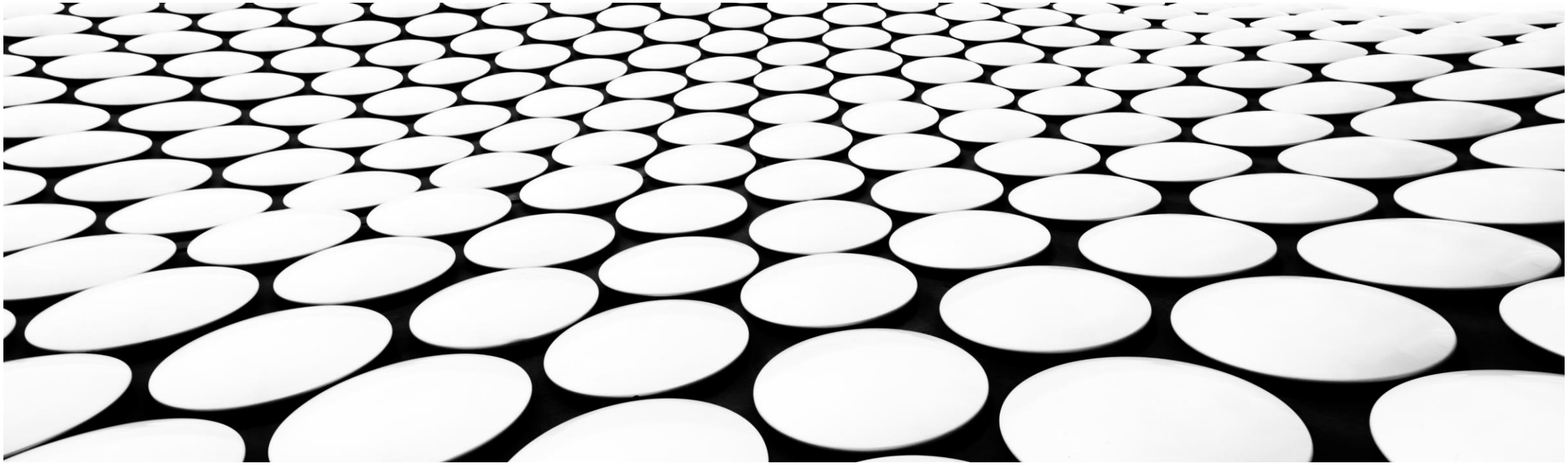
INFO-H-503 – GPGPU Programming – Project

Daniele Bonatto

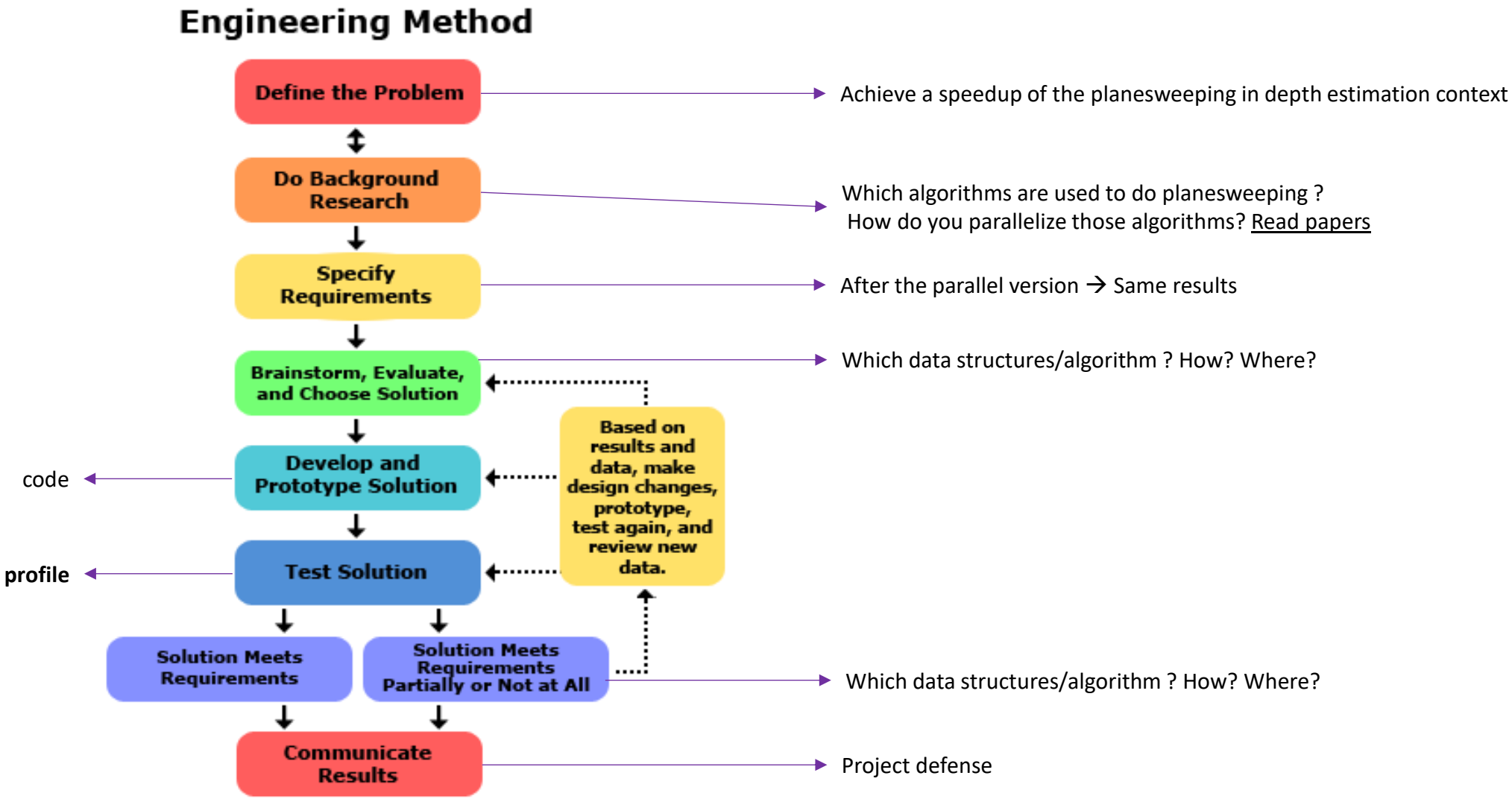
Eline Soetens



Scientific Method

https://www.sciencebuddies.org/science-fair-projects/engineering-design-process/engineering-design-compare-scientific-method

- Follow as much as possible the Engineering Design Process (highly correlated to the scientific method)
- You are not only evaluated on the speed-up but also on how you achieved it

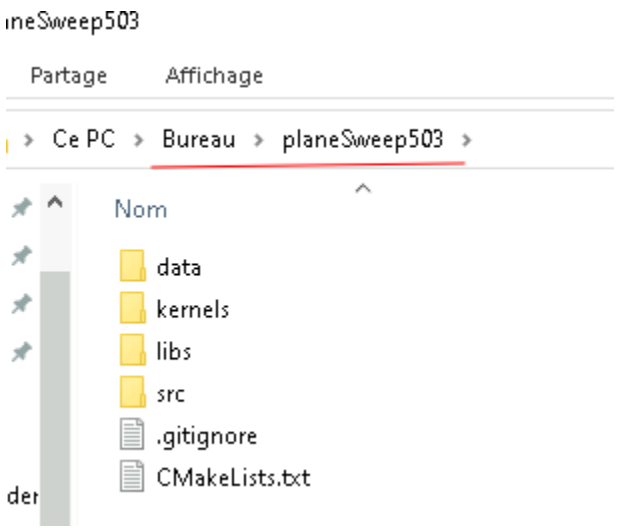


How to compile and START the software

- You need:
  - Visual Studio 2022 (https://aka.ms/vs/17/release/vs_community.exe)
  - Cmake (https://cmake.org/)
  - OpenCV (https://opencv.org/)
  - CUDA 12 installed (https://developer.nvidia.com/cuda-toolkit-archive)
  - Git (https://git-scm.com/)
- The software (PlaneSweep) (given)
- Images (given)
- Camera parameter (given)

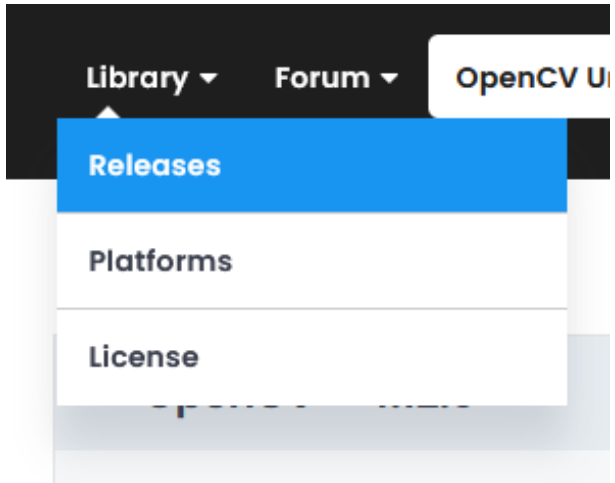
PlaneSweep

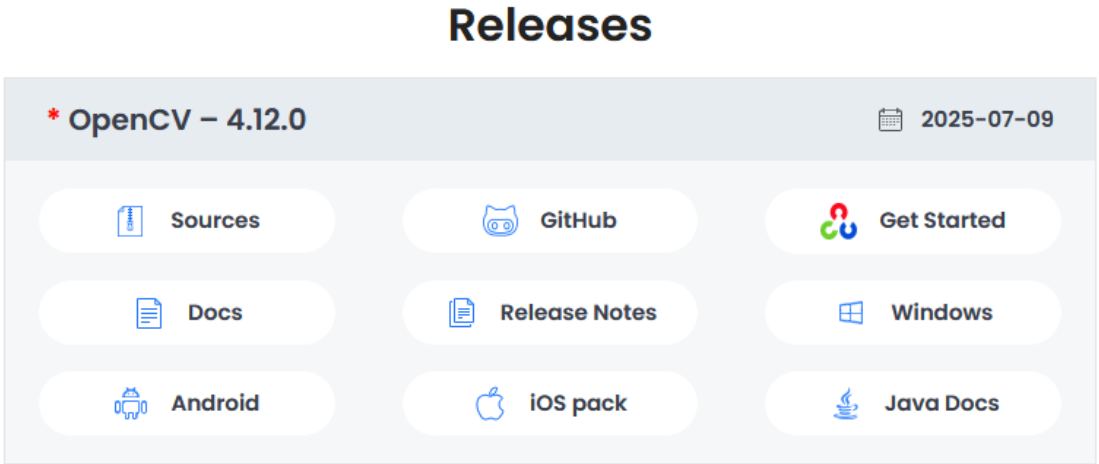
- Download the source code for the project:
- Unzip it in the desktop



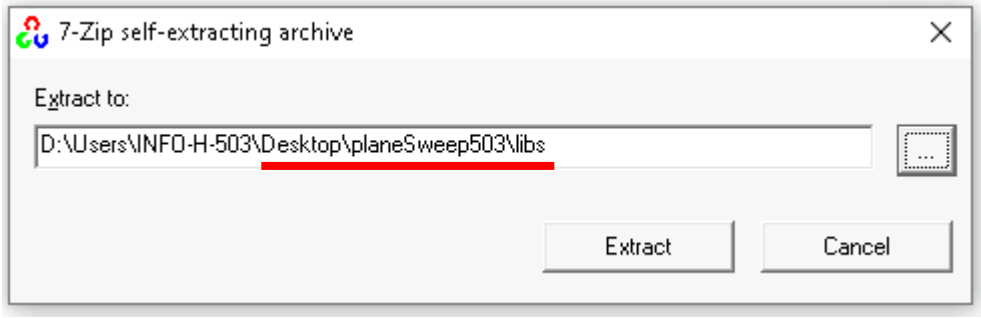
OpenCV

- Go to OpenCV website and download the Windows version





opencv-4.12.0-windows(1).exe

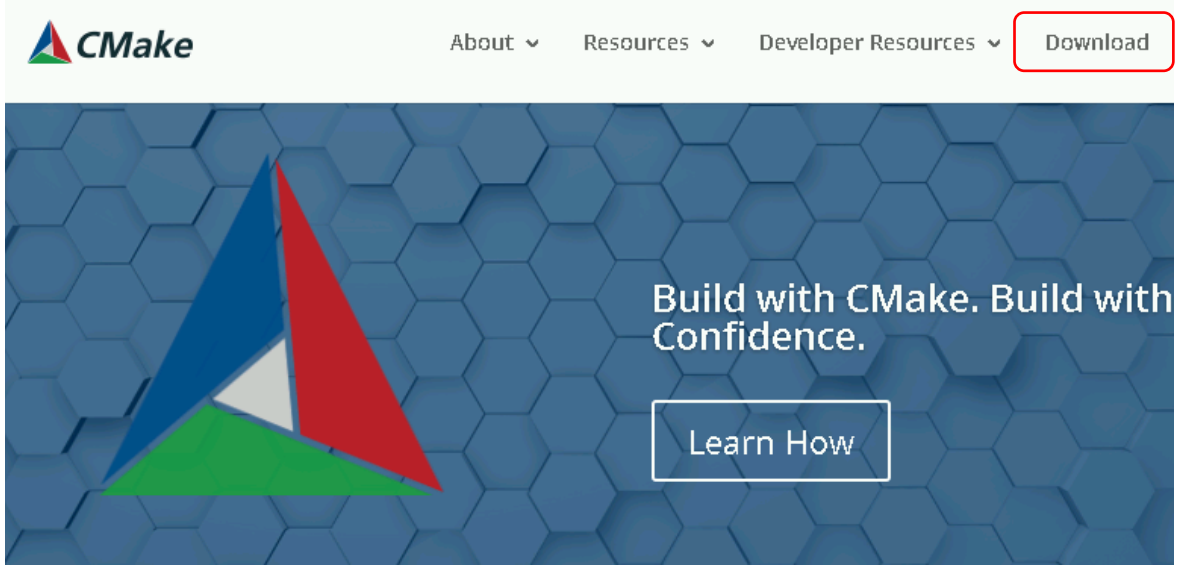


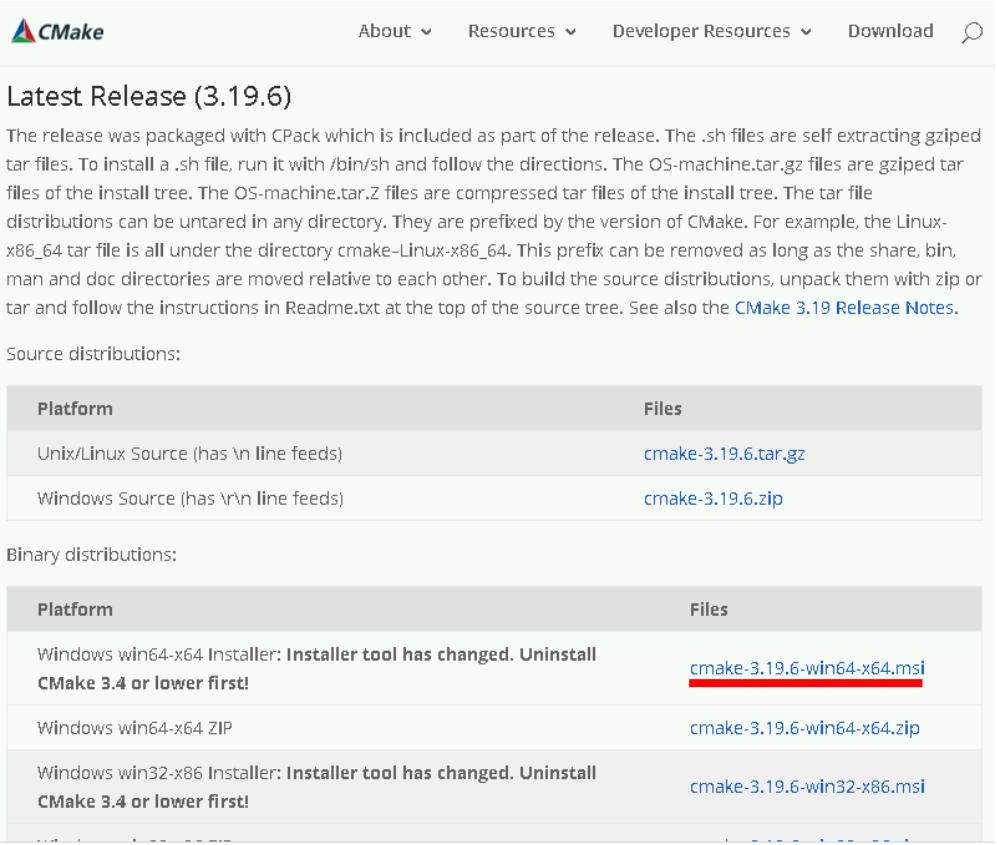
- When you execute the installer, select the subfolder “libs” inside your project

CMAKE

- Download Cmake

(already done on PC from computer room)





CMake About Resources Developer Resources Download

Latest Release (3.19.6)

The release was packaged with CPack which is included as part of the release. The .sh files are self extracting gziped tar files. To install a .sh file, run it with /bin/sh and follow the directions. The OS-machine.tar.gz files are gziped tar files of the install tree. The OS-machine.tar.Z files are compressed tar files of the install tree. The tar file distributions can be untared in any directory. They are prefixed by the version of CMake. For example, the Linux-x86_64 tar file is all under the directory cmake-Linux-x86_64. This prefix can be removed as long as the share, bin, man and doc directories are moved relative to each other. To build the source distributions, unpack them with zip or tar and follow the instructions in Readme.txt at the top of the source tree. See also the CMake 3.19 Release Notes.

Source distributions:

Platform	Files
Unix/Linux Source (has \n line feeds)	cmake-3.19.6.tar.gz
Windows Source (has \r\n line feeds)	cmake-3.19.6.zip

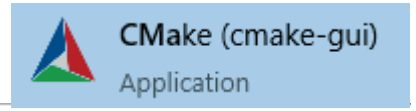
Binary distributions:

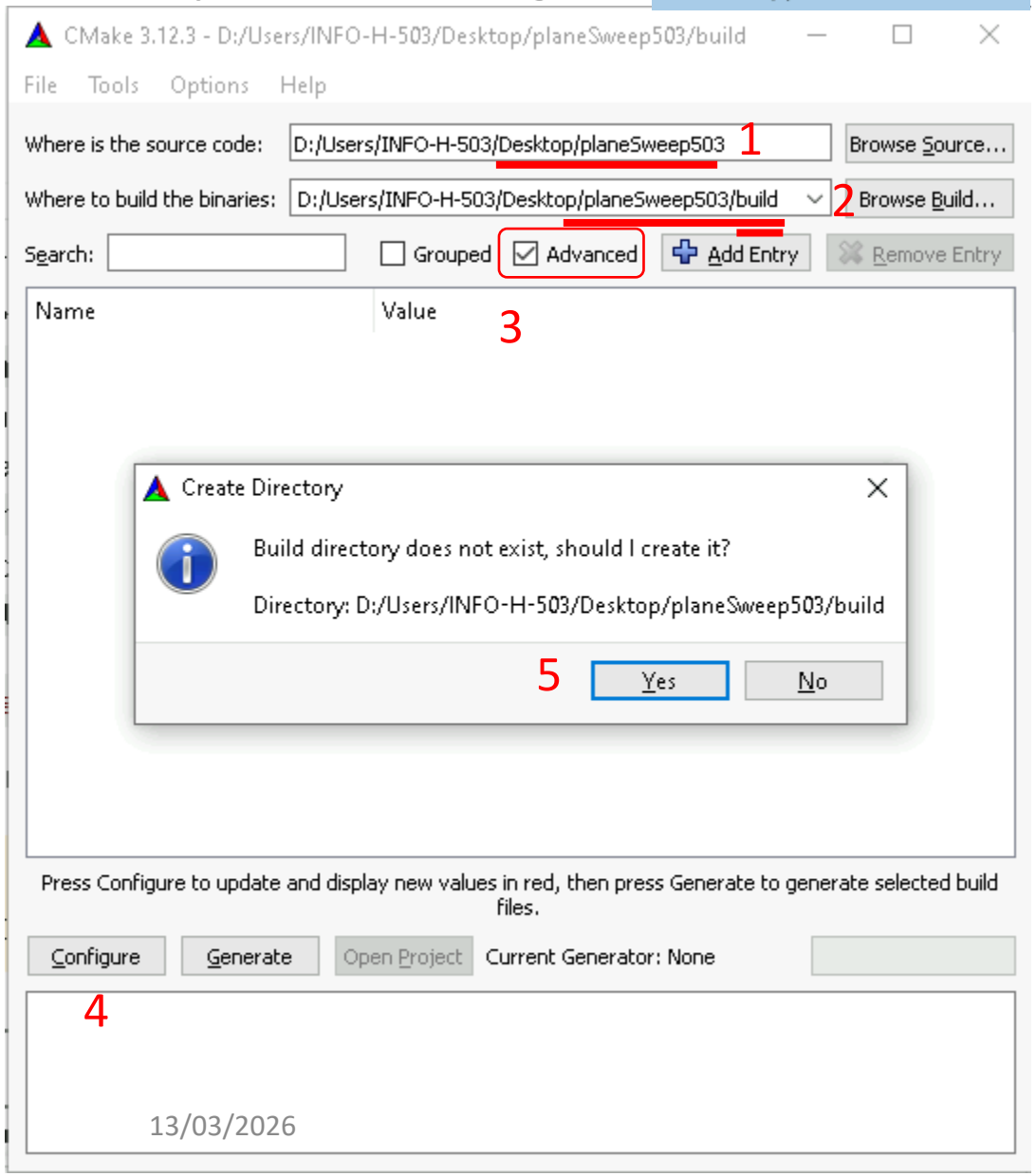
Platform	Files
Windows win64-x64 Installer: **Installer tool has changed. Uninstall CMake 3.4 or lower first!**	cmake-3.19.6-win64-x64.msi
Windows win64-x64 ZIP	cmake-3.19.6-win64-x64.zip
Windows win32-x86 Installer: **Installer tool has changed. Uninstall CMake 3.4 or lower first!**	cmake-3.19.6-win32-x86.msi

- Install Cmake by “next→accept→next-→…→next”

CMAKE

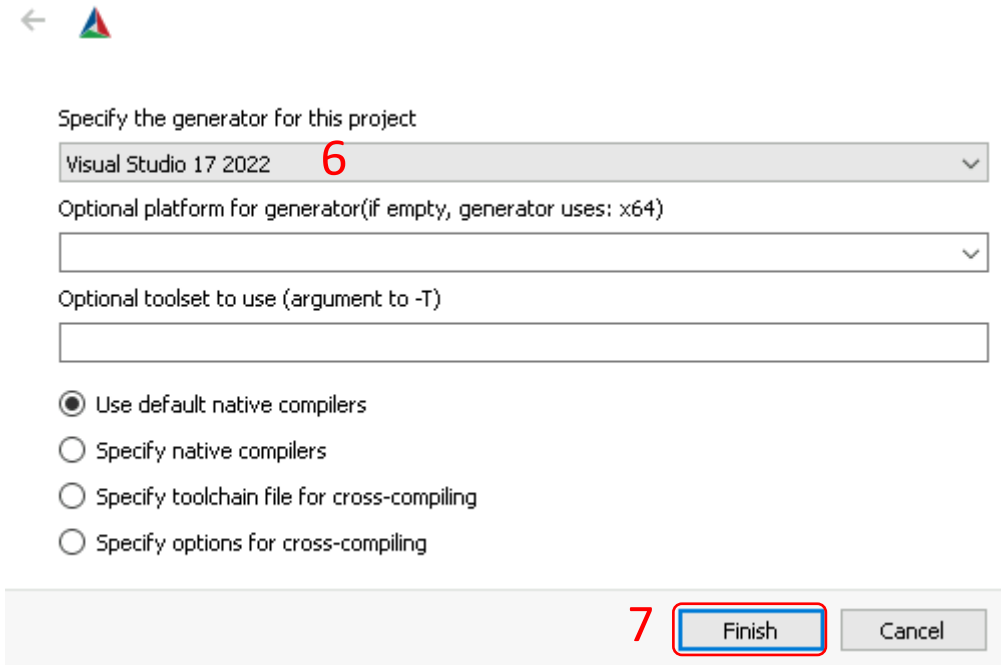
- Open cmake-gui



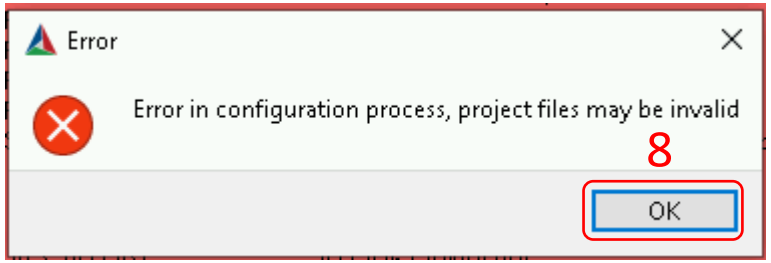


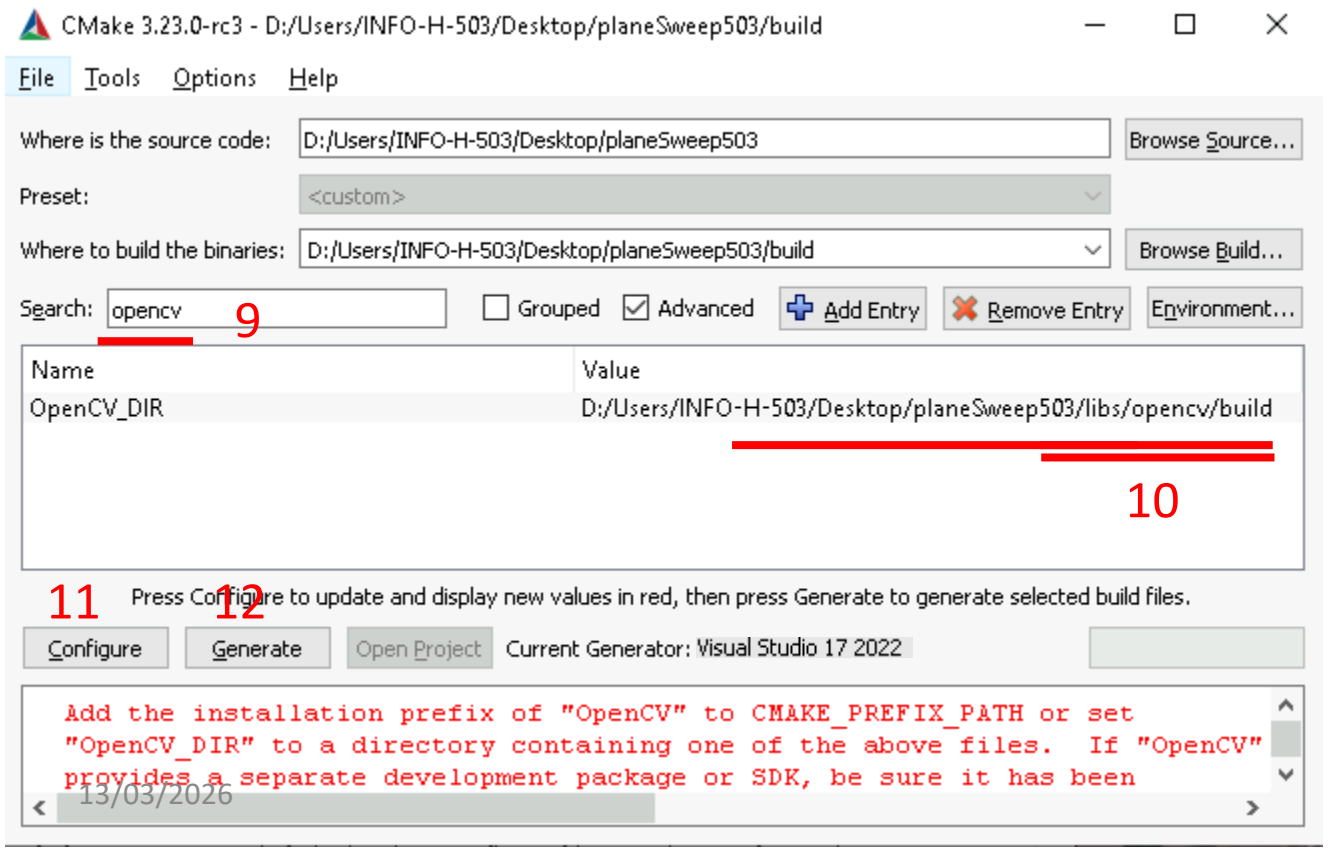
Desktop\project

Desktop\project\build



CMAKE





project\libs\opencv\build

OpenCV

« planeSweep503 › libs › opencv › build › x64 › vc16 › bin

Rechercher dans : bin

Nom	Modifié le	Type	Taille
opencv_annotation.exe	03-07-25 11:05	Application	40 Ko
opencv_interactive-calibration.exe	03-07-25 11:05	Application	139 Ko
opencv_model_diagnostics.exe	03-07-25 11:05	Application	42 Ko
opencv_version.exe	03-07-25 11:05	Application	36 Ko
opencv_version_win32.exe	03-07-25 11:05	Application	34 Ko
opencv_videoio_ffmpeg4120_64.dll	03-07-25 10:50	Extension de l'applica...	27.685 Ko
opencv_visualisation.exe	03-07-25 11:05	Application	54 Ko
opencv_world4120.dll	03-07-25 11:05	Extension de l'applica...	62.712 Ko
opencv_world4120.pdb	03-07-25 11:05	Program Debug Data...	36.412 Ko
opencv_world4120d.dll	03-07-25 10:56	Extension de l'applica...	124.273 Ko
opencv_world4120d.pdb	03-07-25 10:57	Program Debug Data...	270.724 Ko

1. Copy those files

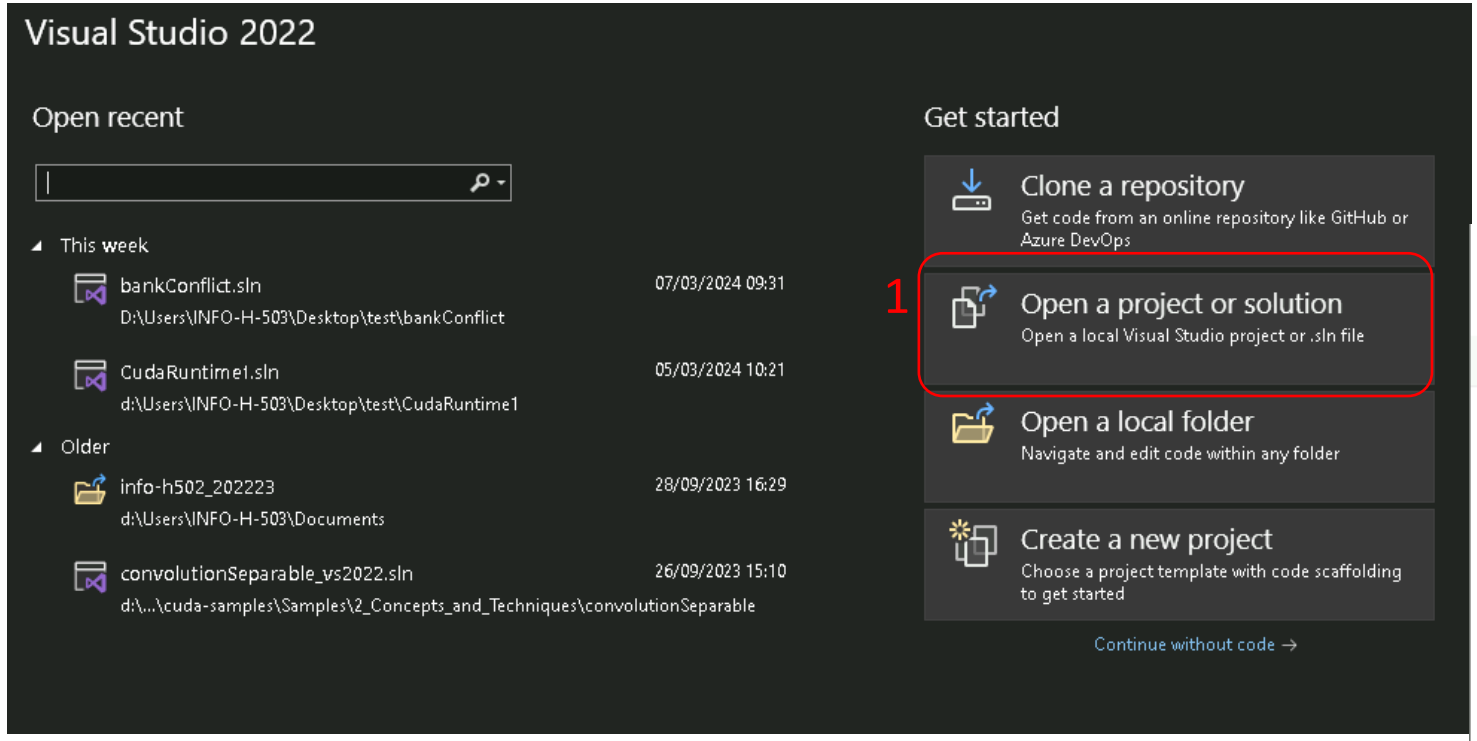
› Ce PC › Bureau › planeSweep503 › build ›

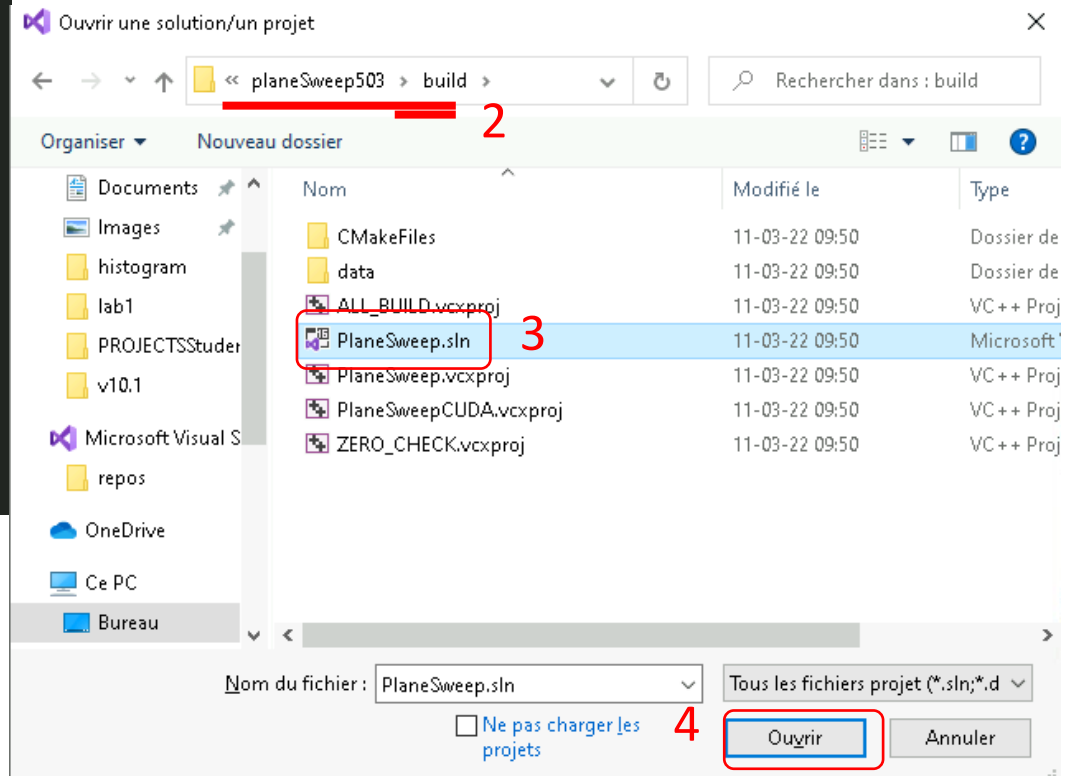
Nom	Modifié le	Type
CMakeFiles	11-03-22 09:50	Dossier de fichiers
data	11-03-22 09:50	Dossier de fichiers
ALL_BUILD.vcxproj	11-03-22 09:50	VC++ Project
ALL_BUILD.vcxproj.filters	11-03-22 09:50	VC++ Project Filte...
cmake_install.cmake	11-03-22 09:50	Fichier CMAKE
CMakeCache.txt	11-03-22 09:50	Document texte
detect_cuda_compute_capabilities.cu	11-03-22 09:50	Fichier CU
PlaneSweep.sln	11-03-22 09:50	Microsoft Visual S...

2. to here

Visual Studio

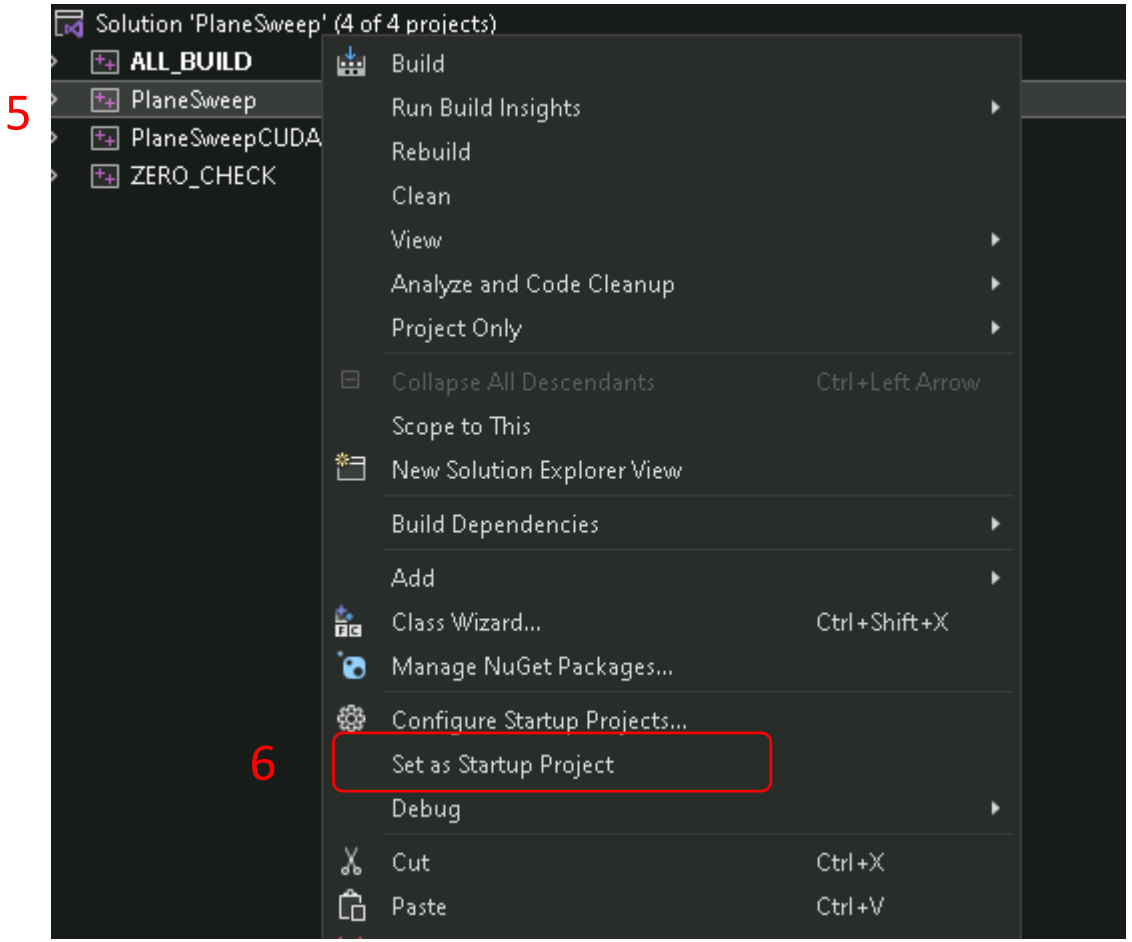
- Open Visual Studio 2022





Visual Studio

- Open Visual Studio 2022



Visual Studio

- Open Visual Studio 2022

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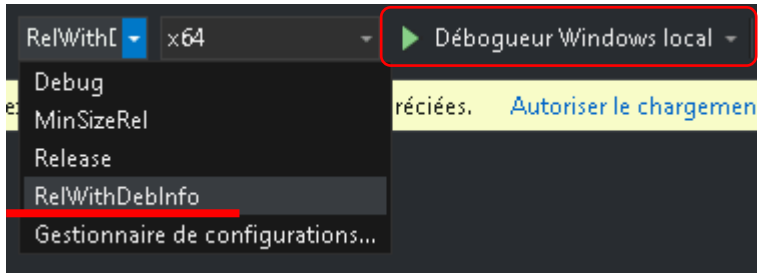
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